

CHAPTER 3 Epidemiologic Measurements Used to Describe Disease Occurrence

Epidemiologic Measures Related to Mortality

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3.10.

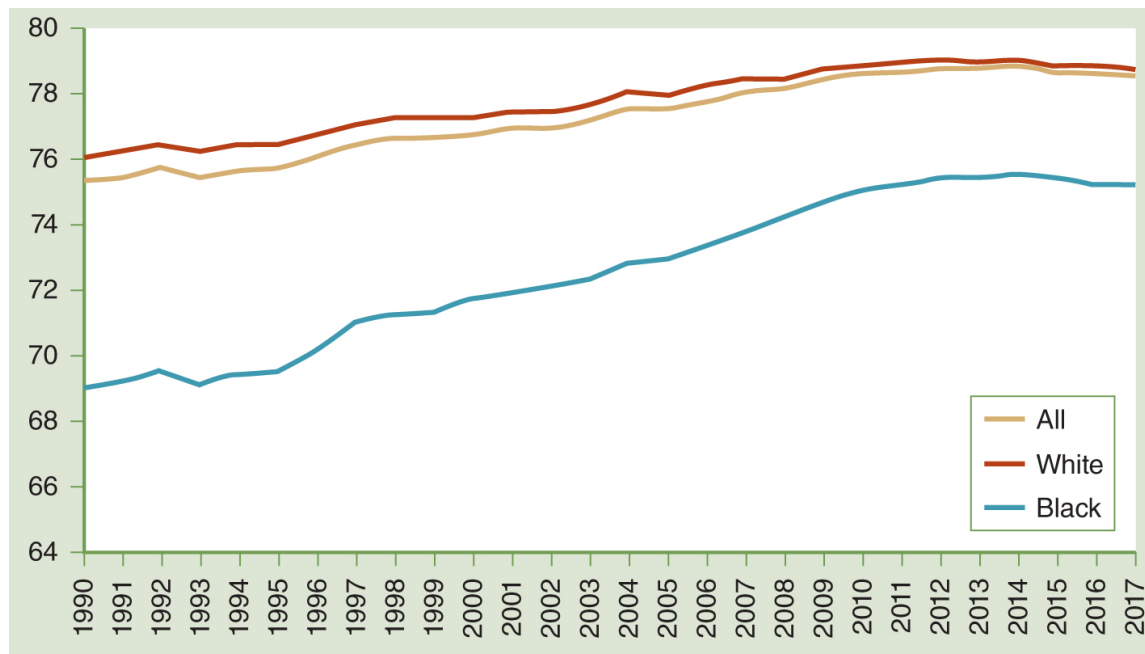


Figure 3.10 Life expectancy at birth in the United States by race, 1990–2017

Data from Arias E, Tejada-Vera B, Kochanek KD, Ahmad FB. *Provisional life expectancy estimates for 2021*. Vital Statistics Rapid Release; no 23. Hyattsville, MD: National Center for Health Statistics; 2022.

Description

Crude Mortality (Death) Rate

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A **crude mortality rate** includes deaths from all causes in the full population of interest. Remember that rates include a time period during which an event occurred. In most cases, for crude mortality rates that time period is 1 year. For the crude mortality rate, the numerator consists of the number of deaths from all causes over a specified time period, and the denominator is a unit size of population from which these deaths occurred.

In the formula shown below, the denominator is also termed the **reference population**, which is defined as the population within which the deaths occurred. The value for the denominator would be the size of the population of the country during that year. The best estimate of the size of a population is often taken as the size of the population around the midpoint of the year, if such information can be obtained. Referring to the formula below, one calculates the U.S. crude mortality rate as 1,020.99 per 100,000 persons for 2020 (the most recently available data as of this writing).

Crude Mortality Rate (e.g., 2020)

$$= \frac{\text{Number of deaths in 2020 from all causes}}{\text{U.S. population at the midpoint of 2020}} \times \text{multiplier}$$

- Multiplier is usually either 1,000 or 100,000.

- *Calculation example:*

In the United States in 2020,

- 3,383,729 deaths from all causes
 - Population as of July 1, 2020 = 331,449,520
- $$\text{Crude mortality rate} = \frac{3,383,729}{331,449,520} \times 100,000$$
- $= 1,020.99 \text{ per } 100,000$

Rates improve our ability to make comparisons, although they also have limitations. For example, we may find that the crude mortality rate in New York is higher than the crude mortality rate in Utah in 2020. Based on this, we might conclude that the risk of death is higher in New York than in Utah. This may be a spurious conclusion, however, because there may be important differences in population composition (e.g., age differences between populations) that would affect mortality experience. Later in this chapter, the procedure to adjust for age differences or other factors is discussed.

Case Fatality Rate

An additional measure covered in this section is the case fatality rate (CFR). The **case fatality rate** refers to the number of deaths due to a disease that occur among people who are afflicted with that disease. The CFR is not actually a rate, but instead is expressed as a percentage and provides a measure of the lethality of a disease. The CFR is defined as the number of deaths due to a specific disease within a specified time period divided by the number of cases of that disease during the same time period multiplied by 100. The formula is expressed as follows:

Case Fatality Rate (CFR)

$$= \frac{\text{Number of deaths due to disease "X"}}{\text{Number of cases of disease "X"}} \times 100$$

- Time period should be specified.

- *Calculation example:*

- 45 hantavirus cases
- 22 deaths from hantavirus
- $\text{CFR} = \frac{22}{45} \times 100 = 48.9\%$

